

HR Attrition Dashboard

Analytical Project Report

Power BI Dashboard | Employee Attrition Analysis | Public CSV Dataset

Prepared for academic and portfolio presentation

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1. Executive Summary

This report documents the HR Attrition Dashboard project created in Power BI. The dashboard converts an employee attrition CSV dataset into interactive reports that help identify workforce patterns, attrition drivers, and retention priorities. The project is appropriate for an academic or portfolio presentation because it uses a public HR analytics dataset instead of private company employee records. This protects confidentiality while still allowing realistic analysis of HR decision-making problems.

The project focuses on a practical business issue: organizations lose time, money, and operational continuity when skilled employees leave. The dashboard supports HR managers by summarizing where attrition is concentrated, which employee groups are vulnerable, and which workplace factors may require intervention.

Project Area	Description
Project Type	Business Intelligence / HR Analytics Dashboard
Platform	Power BI Desktop
Data Source	Public HR Employee Attrition CSV dataset, used to avoid private company data exposure
Dataset Size	1,470 employee records and 35 attributes in the uploaded CSV
Core Target	Analyze employee attrition and support retention strategy
Final Output	Five-page interactive Power BI dashboard with KPIs, charts, filters, and navigation

2. Project Context and Business Problem

Employee attrition is a major HR management problem because it affects productivity, recruitment cost, team stability, and organizational knowledge retention. A company may know how many employees have left, but without analysis it cannot easily understand where the problem is concentrated or what factors are associated with employee exits.

The main business problem for this project is:

How can HR identify the key employee groups and workplace conditions linked with attrition so that management can take targeted retention actions?

The dashboard was created to answer this question visually. It replaces manual inspection of raw rows with a structured set of analytical pages covering demographics, turnover patterns, compensation, overtime, performance, and employee wellness factors.

3. Dataset Source and Ethical Use of Data

The dataset used for this dashboard is a public HR Employee Attrition CSV dataset. It is suitable for academic and portfolio work because it does not expose confidential employee information from a real employer. For presentation purposes, it can be described as a Kaggle-style public HR analytics dataset used to simulate a real-world HR decision-making case.

This approach is ethically safer than using actual company HR data because real employee data may include sensitive information about salary, employment status, performance, and personal demographics. In this project, the public dataset provides realistic fields while avoiding direct harm to any company or individual employee.

4. Dataset Profile

The uploaded CSV contains structured employee-level records. Each row represents one employee, and each column represents a measurable HR attribute such as department, job role, satisfaction rating, income, overtime, years at company, and attrition status.

Metric	Value
Total employee records	1,470
Total attributes / columns	35
Employees who left	237
Employees who stayed	1,233
Overall attrition rate	16.12%
Missing values detected	0
Average age	36.92 years
Average monthly income	6,502.93
Average hourly rate	65.89
Average total working years	11.28 years

Important dataset attributes include:

Column Group	Attributes
Demographics	Age, Gender, MaritalStatus, Education, EducationField, DistanceFromHome, Over18
Employment Profile	Department, JobRole, JobLevel, EmployeeNumber, EmployeeCount
Compensation and Rates	MonthlyIncome, MonthlyRate, DailyRate, HourlyRate, PercentSalaryHike, StockOptionLevel
Work and Career	BusinessTravel, OverTime, TotalWorkingYears, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, Y
Satisfaction and Performance	EnvironmentSatisfaction, JobSatisfaction, RelationshipSatisfaction, JobInvolvement, WorkLifeBalance, PerformanceF
Target Variable	Attrition

5. Methodology and Dashboard Development Process

The project follows a standard BI development process: data collection, data loading, cleaning, modelling, DAX measure creation, dashboard design, and insight interpretation.

Step	Work Performed
1. Data Collection	Downloaded public HR attrition CSV data and stored it locally for Power BI use.
2. Data Loading	Imported the CSV into Power BI using Get Data / Text-CSV.
3. Data Validation	Checked row count, column count, attrition categories, departments, job roles, and missing values.
4. Data Transformation	Prepared categories such as age groups and distance groups where needed for cleaner visual analysis.
5. Data Modelling	Used the employee dataset and a separate measures table for KPI calculations.
6. DAX Measures	Created measures for employee count, active employees, attrition count, attrition rate, gender-level attrition percentages, and av
7. Dashboard Design	Built five report pages with cards, bar charts, donut charts, line charts, and navigation buttons.

8. Optimization	Removed unnecessary images, adjusted layout, improved readability, and kept only relevant dashboard elements.
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6. Key Measures and Calculations

The dashboard uses calculated measures to summarize raw employee records into business indicators. These measures allow the report pages to answer management questions quickly.

Measure	Purpose
Employee Count	Counts the total employees available in the dataset.
Attrition Count	Counts employees whose attrition status is Yes.
Active Employees	Counts employees whose attrition status is No.
Attrition Rate	Shows the percentage of employees who left the organization.
Average Monthly Income	Summarizes employee monthly income level.
Average Hourly Rate	Shows average hourly rate across employees.
Average Working Years	Shows the average total career experience of employees.
Gender Attrition Percentage	Compares attrition distribution between male and female employees.

7. Dashboard Structure

The optimized dashboard contains five pages. Each page has a specific analytical purpose and is connected through navigation buttons at the bottom of the report canvas.

Dashboard Page	Purpose
Overview	Introduces the dashboard and explains the purpose of the four analytical report areas.
Demographics	Analyzes employee attrition by gender, age group, education field, distance from home, work-life balance, and marital status.
Turnover Analysis I	Focuses on attrition by job role, department, business travel, and years in current role.
Turnover Analysis II	Explores attrition by income, job level, overtime, age, and performance rating.
Employee Wellness	Examines satisfaction, performance, work-life balance, job involvement, and wellness-related attrition factors.

8.1 Overview Page



Figure 1: Optimized Power BI dashboard - Overview Page.

Purpose

The Overview page acts as the landing page and navigation hub. It introduces the four major analytical sections: Demographics, Turnover Analysis I, Turnover Analysis II, and Employee Wellness.

What was changed or optimized

The page has a cleaner structure with a purple background, large central title, report icons, and short descriptions. It reduces unnecessary clutter and makes the dashboard easier to understand before entering detailed analysis.

What the visuals represent

Each block explains one report section. The Demographics block represents employee profile analysis; Turnover Analysis I represents role, department, travel, and tenure analysis; Turnover Analysis II represents salary, overtime, age, and performance analysis; Employee Wellness represents satisfaction and work-life balance analysis.

8.2 Demographics Page

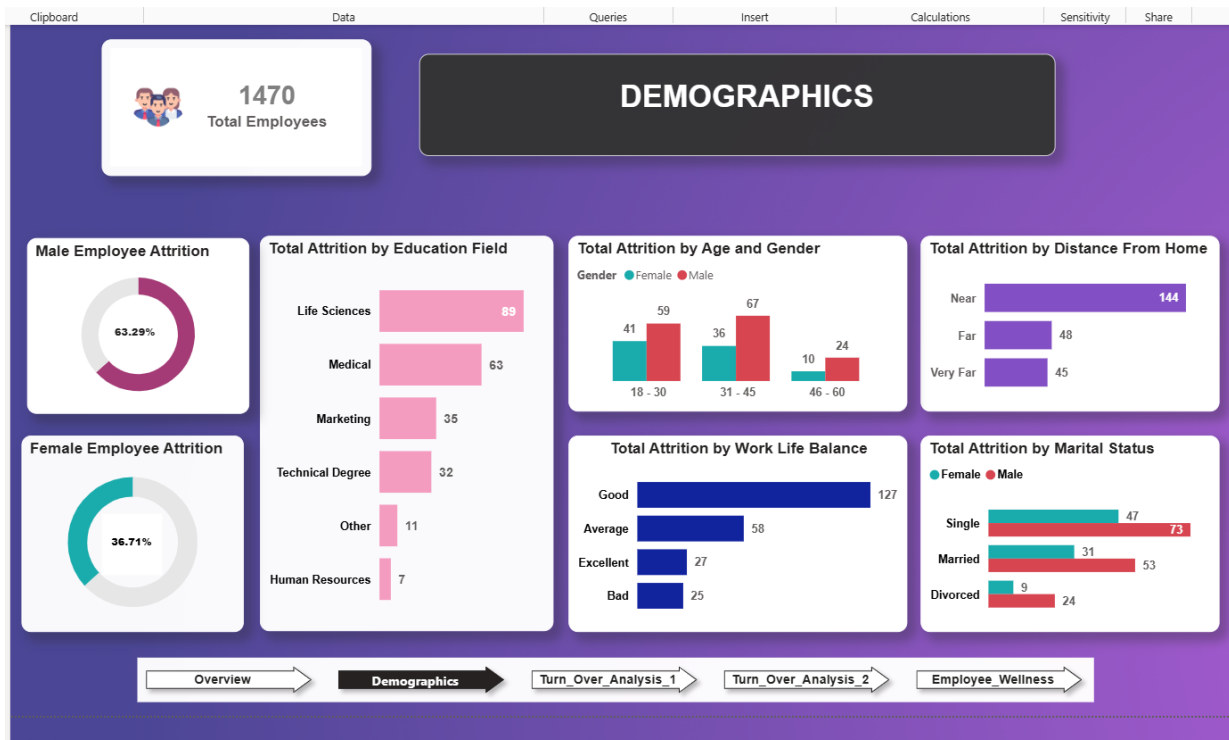


Figure 2: Optimized Power BI dashboard - Demographics Page.

Purpose

This page explains how attrition changes across employee demographic groups.

Charts and meaning

The total employees card shows 1,470 employee records. The male and female donut charts show gender-level contribution to attrition. The education field bar chart shows which educational backgrounds have the highest attrition count. The age and gender clustered bars compare attrition across age groups and gender. The distance from home bar chart shows whether commuting distance is linked with attrition. The work-life balance chart shows attrition counts by balance rating. The marital status chart compares attrition across single, married, and divorced employees.

Main insight

Higher attrition appears in the 18-30 and 31-45 age groups, and the dataset shows larger attrition contribution from male employees. Life Sciences and Medical education fields contribute the highest attrition counts.

8.3 Turnover Analysis I Page

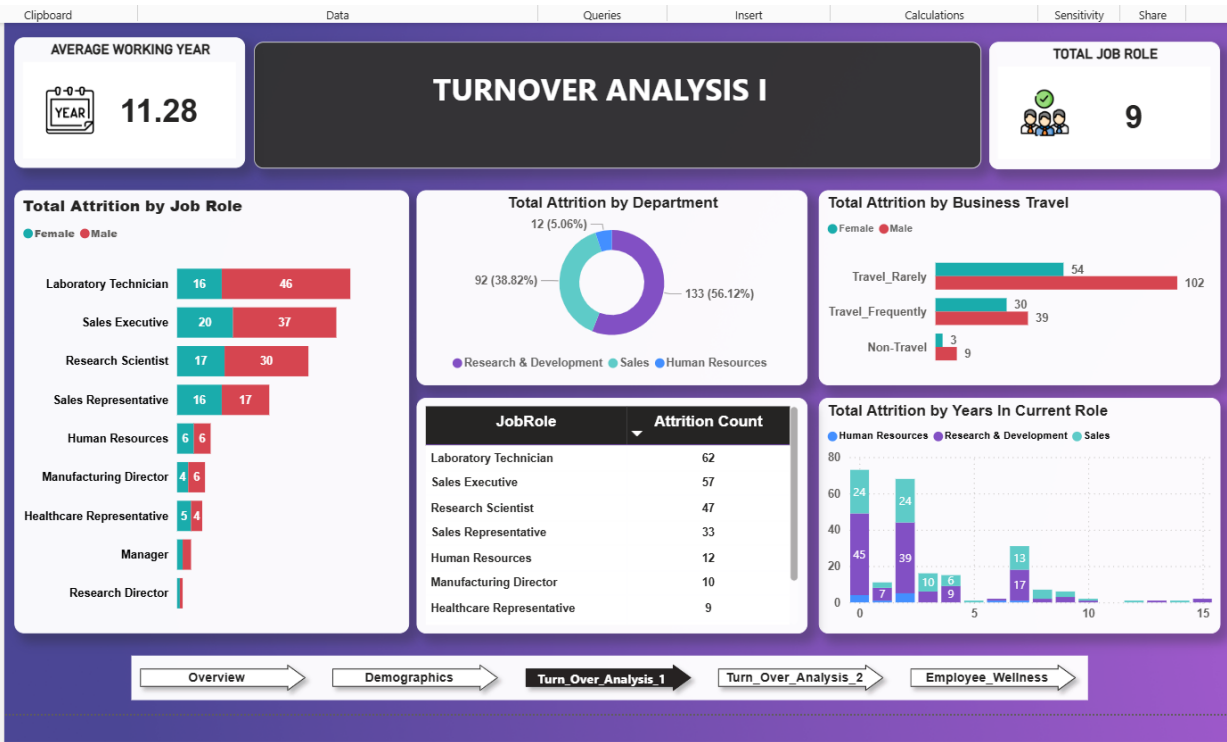


Figure 3: Optimized Power BI dashboard - Turnover Analysis I Page.

Purpose

This page identifies where employee turnover is concentrated by job role, department, travel pattern, and years in current role.

Charts and meaning

The average working year card shows workforce experience level. The total job role card shows nine job roles. The job role bar chart and table rank roles by attrition count. The department donut chart compares turnover concentration across Research & Development, Sales, and Human Resources. The business travel chart compares attrition for rarely traveling, frequently traveling, and non-traveling employees. The years in current role chart shows how tenure in the same role relates to attrition.

Main insight

Laboratory Technician, Sales Executive, and Research Scientist are the most affected job roles. Research & Development has the highest attrition count, followed by Sales.

8.4 Turnover Analysis II Page

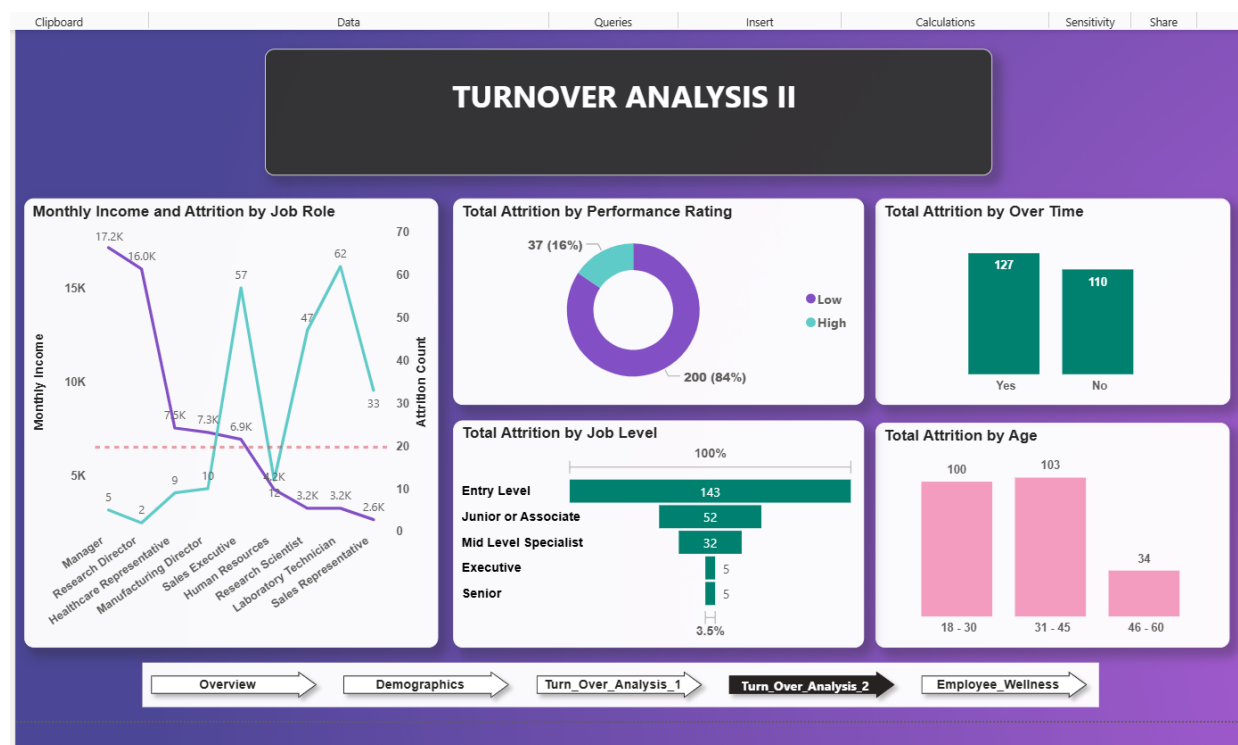


Figure 4: Optimized Power BI dashboard - Turnover Analysis II Page.

Purpose

This page analyzes financial, performance, overtime, job level, and age-related attrition drivers.

Charts and meaning

The line chart compares monthly income and attrition count by job role. The performance rating donut chart compares attrition by low and high performance rating categories. The overtime chart shows attrition split between employees working overtime and not working overtime. The job level funnel/bar chart shows attrition by career level. The age chart shows attrition across the 18-30, 31-45, and 46-60 groups.

Main insight

Entry-level employees have the highest attrition count. Overtime employees show a high attrition count, and the 18-30 and 31-45 age groups are the most affected.

8.5 Employee Wellness Page



Figure 5: Optimized Power BI dashboard - Employee Wellness Page.

Purpose

This page focuses on human factors behind attrition, including employee satisfaction, involvement, relationship satisfaction, performance, and work-life balance.

Charts and meaning

The average monthly income card summarizes income level, and the average hourly rate card summarizes hourly pay. Environment satisfaction, relationship satisfaction, and job satisfaction charts show attrition by satisfaction category. The performance rating chart shows attrition by low/high performance category. The work-life balance donut chart shows attrition by balance level. The job involvement bar chart shows attrition across very low, low, moderate, and high involvement.

Main insight

Wellness metrics help HR understand whether attrition is connected to poor satisfaction, weak engagement, or work-life imbalance. The page supports retention planning beyond salary and job role analysis.

9. Results and Findings

The project successfully converts the raw HR CSV dataset into a readable and interactive Power BI dashboard. The final dashboard provides a complete view of employee attrition across demographic, organizational, financial, and wellness dimensions.

- The dataset includes 1,470 employees and 35 attributes in the uploaded CSV.
- Total attrition count is 237, producing an overall attrition rate of 16.12%.
- Research & Development has the highest attrition count with 133 employees, followed by Sales with 92 employees.
- The highest attrition job roles are Laboratory Technician (62), Sales Executive (57), and Research Scientist (47).
- Entry-level employees and younger/mid-career employees show higher attrition concentration than senior groups.
- Overtime, satisfaction, work-life balance, and job involvement are useful indicators for HR retention decisions.

10. HR Recommendations

Based on the dashboard findings, HR teams can use the following actions to reduce attrition and improve retention.

Problem Area	Recommended Action
High attrition in specific job roles	Create role-specific career paths, internal mobility opportunities, and skill development plans.
High attrition in younger employees	Offer mentoring, transparent promotion criteria, and early-career growth programs.
Overtime-related attrition	Review workload allocation, team capacity, and manager-level planning practices.
Satisfaction-related attrition	Run regular pulse surveys and act on environment, relationship, and job satisfaction issues.
Entry-level attrition	Improve onboarding, buddy programs, feedback frequency, and first-year retention tracking.
Commuting and distance concerns	Consider flexible work arrangements where business operations allow it.

11. Limitations

The dataset is useful for dashboard practice and HR analytics demonstration, but it should not be presented as confidential data from a specific real company. It is a public dataset used for educational analysis. Some important real-world HR factors, such as manager feedback, interview notes, recruitment source, exact resignation reason, and employee survey comments, are not available. Therefore, the dashboard should be used as a decision-support tool, not as the only basis for HR decisions.

12. GitHub Documentation and Upload Plan

To upload the project to GitHub professionally, the repository should clearly include the dashboard file, dataset, screenshots, and a README file explaining the project objective, dataset source, tools, dashboard pages, insights, and limitations.

Repository File	Purpose
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HR_Dashboard.pbix	Main Power BI project file.
HR-Employee-Attrition.csv or .xlsx	Dataset used in the dashboard.
Dashboard screenshots folder	Images of the five dashboard pages.
README.md	Project explanation, dataset details, tools, key insights, and instructions.
Report PDF	Formal report for submission or portfolio presentation.

13. Conclusion

The HR Attrition Dashboard demonstrates how Power BI can be used to convert employee data into practical HR intelligence. The project addresses a real business problem: understanding and reducing employee attrition. By using a public CSV dataset, the project avoids private data exposure while still showing realistic HR analytics skills. The optimized dashboard provides clear pages for demographics, turnover analysis, income and overtime factors, and employee wellness indicators.

The final result is a portfolio-ready Power BI project that shows data loading, data modelling, DAX measure creation, dashboard design, business interpretation, and reporting. The project can be improved further by adding predictive attrition scoring, deeper drill-through pages, department-level filters, and a more detailed data dictionary.

Appendix A: Data Dictionary Summary

The following list summarizes the columns found in the uploaded CSV dataset.

No.	Attribute	No.	Attribute	No.	Attribute
1	Age	13	HourlyRate	25	PerformanceRating
2	Attrition	14	JobInvolvement	26	RelationshipSatisfaction
3	BusinessTravel	15	JobLevel	27	StandardHours
4	DailyRate	16	JobRole	28	StockOptionLevel
5	Department	17	JobSatisfaction	29	TotalWorkingYears
6	DistanceFromHome	18	MaritalStatus	30	TrainingTimesLastYear
7	Education	19	MonthlyIncome	31	WorkLifeBalance
8	EducationField	20	MonthlyRate	32	YearsAtCompany
9	EmployeeCount	21	NumCompaniesWorked	33	YearsInCurrentRole
10	EmployeeNumber	22	Over18	34	YearsSinceLastPromotion
11	EnvironmentSatisfaction	23	OverTime	35	YearsWithCurrManager
12	Gender	24	PercentSalaryHike		